QP CODE: 25900768

Scheme of Valuation

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BBA (HONOURS) REGULAR EXAMINATION NOVEMBER 2025

FIRST SEMESTER (2024 Admission onwards)

Core Course - MG1CCRBBA102 - BUSINESS STATISTICS AND LOGIC

Part A - Answer **any 5** Questions. Each question carries **2** marks

1. Definition of harmonic mean or its applicability – 2 marks.
2. Definition or correct example of bar diagram – 2 marks.
3. Definition of simple regression or equation of *y on x* or *x on y* – 2 marks.
4. Correlation is weak and positive – 1 mark each.
5. Definition or any one measure of skewness – 2 marks.
6. Definition or any one example of equally likely events – 2 marks.
7. Definition or any one example of a progression – 2 marks.
8. Statement of the relationship: Grand parent – grand child – 2 marks.

[5 x 2 = 10]

Part B - Answer **any 4** Questions. Each question carries **5** marks

9. Q1 = 41.33 and Q3 = 76 – 3 marks, Inter Quartile Range = 34.67 – 2 marks,
   QD = 17.33 – 1 mark. [Maximum 5 marks]
   Formula of quartiles, inter quartile range and quartile deviation only – 1 mark each.
10. Definition of arithmetic mean – 2 marks,
    Average number of new customers in A = 52.4 – 2 marks,
    Average number of new customers in B = 47.6 – 2 marks. [Maximum 5 marks]
11. Brief description about the calculation of range – 3 marks,
    Range = 50 – 1 mark, Coefficient of range = 5/9 = 0.5555 – 1 mark.
    Formula of range and coefficient of range only – 0.5 mark each.
12. Setting of class intervals and determination of frequencies – 1 mark each,
    Final table with proper notations – 1 mark.

Marks	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90	90 – 100
No of students	2	2	4	3	3	1

    Definition of frequency table only – 2 marks.
13. Weighted arithmetic mean = $\frac{1.5\times10 + 0.5\times20 + 2.8\times5 + 10\times2}{10+20+5+2}$ = 1.59 – 5 marks.
    Definition or formula of weighted arithmetic mean only – 2 marks.

14. Expected no of errors: 20×0.3 = 6, 10×0.4 = 4, 40×0.45 = 18 – 0.5 mark each,
Expected number of total errors = 28 – 0.5 mark.

Theory of expectation calculation only – 2 marks.

[4 X 5 = 20]

Part C - Answer **any 2** Questions. Each question carries **20** marks

15. Calculation of mean = 41.4 - 5 marks,
Calculation of variance = 263.04 & standard deviation = 16.2185 - 15 marks,

Formula of mean, standard deviation and variance only – 3 marks each.

16. Calculation of median = 24 – 7 marks,
Calculation of mean deviation about median = 14.775 – 7 marks,
Coefficient of mean deviation = 0.6156 – 6 marks.

Formula of median, mean deviation and coefficient of mean deviation about median only – 3 marks each

17. Derivation of the regression coefficients byx = 0.8×20/40 = 0.4 & bxy = 0.8×40/20 = 1.6 – 5 marks each,
Statement of the regression equation of y on x : y - 14.8 = 0.4 [x - 27.6] & the regression equation of x on y: x - 27.6 = 1.6 [y - 14.8] – 2 marks each,
Probable value of y when x = 12 is, 8.56 and probable value of x when y = 30 is, 51.92 – 3 marks each.

Formula of regression coefficients from correlation – 3 marks each.

18. Definition of scatter diagram – 6 marks, Uses of scatter diagram in business – 2 marks,
Assessment of positive correlation, no correlation and limited degree of correlation using scatter plot – 4 marks each.

[2 X 20 = 40]